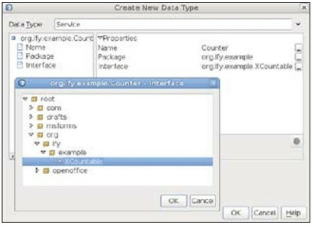

Figure 3: Creating a service

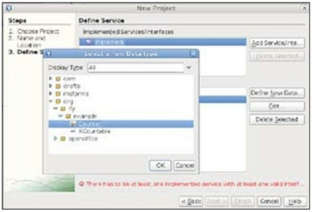

Figure 4: Adding a service

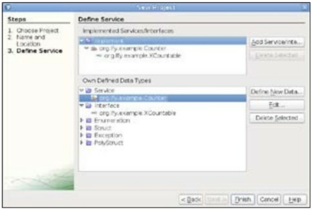

Figure 5: The added service

UNO components

Ref: This is a modified example from the tutorial on Uno/Cpp Component by Daniel Bölzle from the OpenOffice.org wiki. This example is implemented in Java with the NetBeans approach.

A UNO component is an implementation of one or more services provided by UNO or by creating new ones. A service typically wraps interfaces and properties to avail attributes and methods defined in those interfaces.

Here are the steps to create a simple UNO component: *File→New Project→OpenOffice.org→ OpenOffice.org Component.* In the next screen, give a project name, say *SimpleUNO,* with a suitable package name like *org.lfy.example.* The next screen is used to include an existing interface/service or create new ones to be added to the UNO component. In this example we will create a new interface called XCountable with the service, Counter.

The steps to create the XCountable interface are: Select *Interface → Define New Data.* Enter 'Name' as *XCountable.* Now, for the first method, enter *setCount* as 'Name' and *void* as 'Type'. For the parameter in the *setCount* method, enter *nCount* as 'Name' and *long* as 'Type'. Use the 'Next Function' to create the *getCount* method, where we'll enter *getCount* as the 'Name' and *long* as the 'Type'. As we'll not require the 'Parameter' in this case, delete it.

Similarly, create two more methods, *increment* and *decrement*, with no parameters and *long* return type.

The following are the steps to create the counter service: select *Service→Define New Data.* Enter *Counter* as the 'Name' and *org.lfy.XCountable* as the 'Interface'.

Here are the steps to add the counter service: use the *Add Service/Interface* option and navigate to *org→ lfy→example→Counter* to add it.

After all the steps are done, the wizard will look like what's shown in Figure 5.

Now click on the *Finish* button to complete the creation of the project. Observe the created IDL files *XCountable.idl* and *Counter.idl* in the project explorer to get a better idea about IDL files.

We now need to implement methods defined in the XCountable interface. Open *SimpleUNO.java* from the project navigator as follows: *SimpleUNO→Source Packages→ org.lfy.example→ SimpleUNO.java*—change these four methods as shown below and add a global variable, say, *xCount* to be used by all four methods:

```
long xCount;
public int getCount()
{
        return xCount;
}
public void setCount(int nCount)
{
        xCount=nCount;
}
public int increment()
{
        xCount++;
        return xCount;
}
public int decrement()
{
```